Finite PR-box simulation of partially entangled qubits

Abstract

For every fixed partially entangled pure two-qubit state, we give an exact simulation of all local projective-measurement correlations using a fixed finite number of binary PR boxes. The construction allows Borel local wirings and unrestricted measurement-independent shared randomness. An exact finite-list sampling lemma yields an explicit, though quantitatively loose, worst-case bound on the number of boxes.

Theorem. Fix $0 < \theta < \pi/4$. Under the measure-theoretic model in which local wirings are Borel stochastic maps and measurement-independent shared randomness is unrestricted, there is a finite integer N_θ and a fixed local wiring of N_θ independent binary PR boxes reproducing

$$P_\theta(a, b \mid x, y) = \mathrm{Tr}\Big[|\psi_\theta\rangle\langle\psi_\theta|(M_{a|x} \otimes M_{b|y})\Big]$$

for every $x, y \in S^2$, where

$$|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle, \qquad M_{a|x} = \frac{I + (-1)^a x \cdot \sigma}{2}.$$

The number of boxes is a fixed worst-case bound. One possible choice, with $c = \cos(2\theta)$, is

$$k = \left\lceil \left(\frac{12\pi(1+c)}{1-c} \right)^2 \right\rceil + 1, \qquad N_\theta = k^2. \tag{1}$$

"Fixed wiring" means one preselected map whose arguments include its own local measurement direction. Neither the map, the resource nor its number of copies is chosen afresh as a function of the pair of directions.

The resource restriction is finiteness of the PR-box alphabet and number of uses; it does not impose finite precision or a computational time bound on the Borel local maps or shared randomness. The bound (1) is only a sufficient bound, not a lower bound on the required number of boxes. As θ approaches $\pi/4$ from below it gives $1423^2 = 2{,}024{,}929$ boxes, so it is quantitatively very loose.

A single PR box suffices for projective measurements on a maximally entangled qubit pair [CGMP05], whereas some partially entangled pairs require more than one [BGS05]. Brunner, Gisin, Popescu and Scarani [BGPS08, Section IV] simulate these correlations using PR boxes together with a continuous-input millionaire box, and ask whether finite-input resources suffice. The theorem proves finite binary PR-box sufficiency in the stated Borel model. Its proof uses the local-plus-unbiased decomposition from [RQ23] and the finite-list selection argument below; it does not simulate the millionaire box itself.

1 An exact finite-list sampling lemma

Let S and X be compact metric spaces, let q be a Borel probability on S, and let $f : X \times S \to [0, \infty)$ be continuous with

$$\int_S f(x,t)\, q(dt) = 1, \qquad f(x,t) \le M \tag{2}$$

for every x, t. Suppose S admits a sequence of finite Borel partitions of equal positive q-mass, with mesh tending to zero and cardinality tending to infinity.

Then there is a fixed finite list $Z_1, \dots, Z_N$, drawn independently of x, and a single Borel function

$$I : X \times S^N \times [0,1] \longrightarrow \{1, \dots, N\} \tag{3}$$

such that, with U independent uniform randomness, $Z_{I(x,Z,U)}$ has law $\nu_x(dt) = f(x,t)q(dt)$ for every x.

Proof: the capacity bound. Since each density averages to one, it is at least one somewhere. Uniform continuity on $X \times S$ gives a radius $r > 0$, independent of x, such that each f_x is at least $\epsilon = 1/2$ throughout a ball of radius r. Take an equal-mass partition $C_1, \dots, C_N$ with mesh less than r. For every x, the cell containing a point where $f_x \geq 1$ therefore satisfies

$$f_x \geq \epsilon \quad \text{on that entire cell.} \tag{4}$$

The point need not be selected measurably in x; it is used only to prove the inequality below.

Draw the Z_i independently, with law Nq restricted to C_i. Put $\delta = 1/(2M)$. Choose the partition sufficiently fine and N sufficiently large that

$$1 - e^{-N\delta} \geq \tfrac{1}{2}, \qquad e^{\,1-N\delta} \leq \frac{\epsilon}{N}. \tag{5}$$

These conditions are possible along the partition sequence.

For any Borel $B \subseteq S$, set

$$t = q(B), \quad \beta_i = Nq(B \cap C_i), \quad H(B) = \Pr\{\exists i : Z_i \in B\} = 1 - \prod_i (1 - \beta_i).$$

Here $0 \leq \beta_i \leq 1$ and $\sum_i \beta_i = Nt$. If $t \leq \delta$, AM–GM and concavity give

$$\begin{aligned} H(B) &\geq 1 - (1-t)^N \geq 1 - e^{-Nt} \\ &\geq \frac{t}{\delta}(1 - e^{-N\delta}) \geq Mt \geq \nu_x(B). \end{aligned} \tag{6}$$

If $t \geq \delta$, choose a cell C_d satisfying (4). Then

$$\nu_x(B^c) \geq \epsilon q(C_d \setminus B) = \frac{\epsilon(1 - \beta_d)}{N},$$

whereas

$$\begin{aligned} 1 - H(B) &= \prod_i (1 - \beta_i) \\ &\leq (1 - \beta_d) \exp\left(- \sum_{i \neq d} \beta_i \right) \\ &\leq (1 - \beta_d) e^{1-N\delta} \leq \frac{\epsilon(1 - \beta_d)}{N} \leq \nu_x(B^c). \end{aligned} \tag{7}$$

No division by $1 - \beta_d$ is used, so the case $\beta_d = 1$ is included. Thus, for every x and every Borel set,

$$\nu_x(B) \leq H(B). \tag{8}$$

Proof: sufficiency. This is the finite-list instance of the random-set selection criterion of Strassen [S65, Theorem 5] and Artstein [A83, Theorems 3.1–3.2]. We give the compact-space

argument, including the additional parameter measurability. Write $\Omega = S^N$, let K be the fixed list law, and put

$$E = \{(z_1, \dots, z_N; t) : t = z_i \text{ for at least one } i\}.$$

The relation E is closed in the compact metric space $\Omega \times S$. The probability measures π on this space with $\pi_\Omega = K$ and $\pi(E) = 1$ form a nonempty compact convex set. Nonemptiness follows, for example, by selecting z_1. Compactness follows from compactness of the space of probability measures and the closed constraints; for the support constraint, use Portmanteau on the closed E. Their second marginals form a compact convex set Γ.

For $h \in C(S)$,

$$\sup_{\eta \in \Gamma} \int h \, d\eta = \int_\Omega \max_i h(z_i) \, K(dz). \tag{9}$$

The upper bound is pointwise; equality uses the first index maximizing $h(z_i)$, a Borel choice. After adding a constant, assume $h \geq 0$. Apply (8) to the sets $\{h > t\}$ and integrate over t. Layer-cake integration yields

$$\int h \, d\nu_x \leq \int_\Omega \max_i h(z_i) \, K(dz). \tag{10}$$

The same holds for every real $h \in C(S)$. If $\nu_x \notin \Gamma$, Hahn–Banach separation in the weak topology of finite measures would give a continuous h contradicting (9)–(10). Therefore a coupling with second marginal ν_x exists.

Proof: one Borel selection kernel for all x. The map $x \mapsto \nu_x$ is weakly continuous by (2) and uniform continuity of f. Consequently

$$G = \{(x, \pi) : \pi_\Omega = K, \ \pi_S = \nu_x, \ \pi(E) = 1\} \tag{11}$$

is compact, with nonempty compact fibers G_x, in $X \times \mathcal{P}(\Omega \times S)$. These are compact metric spaces.

Here is an explicit measurable-selection argument for these exact hypotheses. Take a countable dense sequence t_j in $\mathcal{P}(\Omega \times S)$. Recursively choose the smallest $j_n(x)$ for which G_x, intersected with all previously chosen closed balls, intersects $\overline{B}(t_{j_n(x)}, 2^{-n})$. Some such index exists by density. For any fixed finite history and any candidate index, this test is a projection of a compact subset of G, hence is closed in X. Countably partitioning according to the histories and taking the first successful index shows that each j_n is Borel. The resulting nested compact intersections have diameters at most 2^{1-n}. The centers $t_{j_n(x)}$ converge to their unique common point $\pi_x \in G_x$; pointwise limits into a metric space are Borel. Thus $x \mapsto \pi_x$ is Borel.

On E assign $J(z, t)$ to the first index with $t = z_i$, and define the finite subprobability kernels

$$\mu_i^x(D) = \pi_x\{(z, t) \in E : z \in D, \ J(z, t) = i\} \quad (D \subseteq \Omega \text{ Borel}). \tag{12}$$

The sets in (12) are Borel. Evaluation of a Borel set by a probability measure is a Borel function of that measure, so these are Borel kernels in x. They satisfy

$$0 \leq \mu_i^x \leq K, \qquad \sum_i \mu_i^x = K. \tag{13}$$

To obtain jointly Borel Radon–Nikodym densities, take refining finite Borel partitions $\mathcal{D}_n$ of Ω generating its Borel sigma algebra. Such partitions are the atoms generated by the first n sets of a countable base. On an atom D of positive K-mass, define

$$w_i^n(x, z) = \frac{\mu_i^x(D)}{K(D)} \quad (z \in D); \tag{14}$$

put zero on zero-mass atoms. These functions are jointly Borel and lie in $[0,1]$. For each fixed x, they are the conditional expectations of $d\mu_i^x/dK$ on $\mathcal{D}_n$. Bounded martingale convergence and the generating property imply their almost-sure convergence to that derivative. Take their pointwise limsups, normalize the finite nonnegative vector when its sum is positive, and otherwise use $(1,0,\dots,0)$. The resulting functions $w_i(x,z)$ are jointly Borel, sum to one everywhere, and satisfy

$$\mu_i^x(D) = \int_D w_i(x,z)\, K(dz) \tag{15}$$

for every fixed x and every Borel D.

Use $U \sim \mathrm{Unif}[0,1)$ to choose the first i whose cumulative weight exceeds U; define $I = 1$ at $U = 1$. This is a single Borel rule. For any Borel $B \subseteq S$,

$$\Pr(Z_I \in B \mid x) = \int_\Omega \sum_i w_i(x,z)\mathbf{1}_B(z_i)\, K(dz) = \pi_x(\Omega \times B) = \nu_x(B). \tag{16}$$

This completes the lemma.

The measure π_x is used to define the local rule; it is not the shared prior. The actual list law is always the same K. The independent seed U matters: Artstein's counterexample shows that capacity inequalities need not permit deterministic selection from the original list alone. Exceptional sets for (15) may depend on x. Equation (16) nevertheless holds for every fixed x, and the rule is jointly Borel. No simultaneous uncountable intersection of full-measure sets is being assumed.

2 The known local-plus-unbiased decomposition

Put

$$c = \cos(2\theta), \quad s = \sin(2\theta), \quad z = (0,0,1), \quad w_x = (sx_1, -sx_2, x_3).$$

For sign outcomes $A = (-1)^a, B = (-1)^b$, direct expansion of the state gives

$$P_\theta(A,B \mid x,y) = \frac{1 + Acx_3 + Bcy_3 + AB\, w_x \cdot y}{4}. \tag{17}$$

In particular the x_2y_2 term has coefficient $-s$. Throughout, $\mathrm{sgn}(t) = +1$ for $t \geq 0$ and -1 otherwise. Write $\Theta(t) = \max(t,0)$, and use sphere surface measure $d\omega$.

For $\eta \in \{+1,-1\}$, define

$$r_\eta = \frac{cz + \eta w_x}{2}, \qquad p_\eta = \frac{1 + \eta c x_3}{2}.$$

Since

$$\|cz + \eta w_x\|^2 = (1 + \eta c x_3)^2,$$

we have $\|r_\eta\| = p_\eta \geq (1-c)/2 > 0$. Set

$$\rho_x(\lambda) = \frac{\Theta(r_+ \cdot \lambda) + \Theta(r_- \cdot \lambda)}{\pi}, \qquad T_x(\eta \mid \lambda) = \frac{\Theta(r_\eta \cdot \lambda)}{\pi \rho_x(\lambda)}. \tag{18}$$

At $\rho_x = 0$, define $T_x(\pm 1 \mid \lambda) = 1/2$.

For unit vectors v, y,

$$\int_{S^2} \Theta(v \cdot \lambda)\, d\omega = \pi, \qquad \int_{S^2} \frac{\Theta(v \cdot \lambda)}{\pi}\, \mathrm{sgn}(y \cdot \lambda)\, d\omega = v \cdot y. \tag{19}$$

For the second identity, write $\Theta(t) = (t+|t|)/2$. The absolute-value term times the sign is odd almost everywhere. Rotational symmetry and one-dimensional integration give $\int \lambda \operatorname{sgn}(y \cdot \lambda) d\omega = 2\pi y$. The first identity follows by integrating over a hemisphere.

Thus ρ_x is normalized. Sampling it, producing Alice's output with response T_x, and setting Bob's output to $\operatorname{sgn}(y \cdot \lambda)$ gives joint probabilities

$$p_\eta \frac{1 + B(r_\eta/p_\eta) \cdot y}{2} = \frac{1 + \eta c x_3 + B c y_3 + \eta B\, w_x \cdot y}{4}, \tag{20}$$

which are exactly (17).

Now define

$$\rho_z(\lambda) = \frac{\Theta(z \cdot \lambda)}{\pi}, \qquad \widetilde{\rho}_x(\lambda) = \rho_x(\lambda) - c\rho_z(\lambda).$$

Using $|u+v| + |u-v| = 2\max(|u|, |v|)$ in (18) yields

$$\widetilde{\rho}_x(\lambda) = \frac{\big(|w_x \cdot \lambda| - c|z \cdot \lambda|\big)_+}{2\pi}. \tag{21}$$

It is nonnegative, even and continuous, with integral $1 - c$.

Let L use the input-independent density ρ_z, Alice's response T_x, and Bob's sign response. This is local. Let Q use density $\widetilde{\rho}_x/(1-c)$ and the same responses. Then

$$P_\theta = cL + (1-c)Q. \tag{22}$$

Where $\widetilde{\rho}_x > 0$, $|w_x \cdot \lambda| > c|z \cdot \lambda|$, so exactly one of $r_+ \cdot \lambda, r_- \cdot \lambda$ is positive. Therefore the residual's responses are deterministic:

$$A = \operatorname{sgn}(w_x \cdot \lambda), \qquad B = \operatorname{sgn}(y \cdot \lambda). \tag{23}$$

The density in Q is even, while both responses are odd outside null great circles. Since $s > 0$, $w_x \neq 0$ for every x. Consequently both residual marginals are exactly uniform, for every x, y. In particular Q is nonsignalling and is determined by

$$C_Q(x,y) = \int_{S^2} \frac{\widetilde{\rho}_x(\lambda)}{1-c} \operatorname{sgn}(w_x \cdot \lambda) \operatorname{sgn}(y \cdot \lambda)\, d\omega. \tag{24}$$

The local weight $c = \cos(2\theta)$ appears in the earlier local-content analysis of Portmann, Branciard and Gisin [PBG12], whose angle convention uses $\theta/2$ in the state. The analytic Kochen–Specker construction used here is from Renner–Quintino [RQ23, Appendices C and E], with $p = \cos^2\theta$ and $2p - 1 = c$. The calculation above records its marginals in the present conventions.

3 A fixed finite list for the residual

Let $q = d\omega/(4\pi)$ and define the normalized density

$$f_x(\lambda) = \frac{2}{1-c}\big(|w_x \cdot \lambda| - c|z \cdot \lambda|\big)_+. \tag{25}$$

It is jointly continuous on $S^2 \times S^2$, bounded by $M = 2/(1-c)$, and is Lipschitz in λ for Euclidean distance with constant

$$L_0 = \frac{2(1+c)}{1-c}. \tag{26}$$

Thus any point where $f_x \geq 1$ has a neighborhood of radius $1/(2L_0)$ on which $f_x \geq 1/2$, uniformly in x.

An explicit equal-area partition is obtained by splitting the height coordinate into k intervals of length $2/k$ and longitude into k intervals of length $2\pi/k$. Off the poles, sphere area is $d\phi\, dh$, so each cell has q-mass $1/k^2$. Use half-open intervals, with the final height endpoint included, and assign each pole to a cell in its adjacent height band. This gives a disjoint Borel partition of the entire sphere.

Its geodesic mesh is at most

$$\frac{\pi}{\sqrt{k}} + \frac{2\pi}{k} \leq \frac{3\pi}{\sqrt{k}}. \tag{27}$$

Indeed, for polar angles $\alpha, \beta \in [0, \pi]$,

$$|\cos\alpha - \cos\beta| = 2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{|\alpha-\beta|}{2}\right) \geq 2\sin^2\left(\frac{|\alpha-\beta|}{2}\right).$$

Height difference at most $2/k$ therefore gives polar-angle difference at most $2\arcsin(1/\sqrt{k}) \leq \pi/\sqrt{k}$. Joining points by a meridian and a parallel adds at most their longitude difference $2\pi/k$. The pole assignments obey the same bound.

Choose k, N as in (1). Then (27) is strictly less than $(1-c)/(4(1+c)) = 1/(2L_0)$, so every x has an entire positive cell, including its assigned boundary points. With $\epsilon = 1/2$ and $\delta = (1-c)/4$, this choice also satisfies (5). To check this explicitly, $k \geq 8$ and $k\delta > 36\pi^2(1+c)^2/(1-c) > 1$, whence $N\delta \geq \sqrt{N}$. For $N \geq 64$, $\sqrt{N} \geq 1 + \log(2N)$: it holds at 64 and the difference has positive derivative for $N > 4$. Hence

$$e^{1-N\delta} \leq \frac{1}{2N}, \qquad 1 - e^{-N\delta} \geq \frac{1}{2}.$$

The lemma supplies a fixed Borel index rule $I(x, \lambda_1, \ldots, \lambda_N, U)$ whose selected point has exactly density (25).

4 The finite nonsignalling wiring

The common resource R is the ordinary PR box:

$$R(\alpha, \beta \mid u, v) = \begin{cases} \frac{1}{2}, & \alpha \oplus \beta = uv, \\ 0, & \text{otherwise}, \end{cases} \qquad u, v, \alpha, \beta \in \{0, 1\}. \tag{28}$$

Its local marginals are fair and input-independent. Use $N = N_\theta$ independent copies. The same binary box can be used for every θ.

Share in advance the stratified list $(\lambda_1, \ldots, \lambda_N)$, an independent uniform U, an independent point λ_0 with density ρ_z, a uniform seed V for Alice's stochastic local response, and a branch bit with local-branch probability c. All these have one fixed joint law independent of x, y, and are independent of the PR resources.

In the local branch, Alice uses V to sample $T_x(\cdot \mid \lambda_0)$; Bob outputs $\mathrm{sgn}(y \cdot \lambda_0)$. They input $u_i = v_i = 0$ into all N boxes and ignore their outputs. This branch gives L.

In the residual branch Alice computes

$$I = I(x, \lambda_1, \ldots, \lambda_N, U), \qquad d = \frac{1 - \mathrm{sgn}(w_x \cdot \lambda_I)}{2}.$$

Bob computes the N bits

$$e_i = \frac{1 - \operatorname{sgn}(y \cdot \lambda_i)}{2}.$$

They query all boxes in parallel, using only their own local inputs:

$$u_i = \mathbf{1}_{\{I=i\}}, \qquad v_i = e_i. \tag{29}$$

Their final output bits are

$$a = d \oplus \bigoplus_{i=1}^{N} \alpha_i, \qquad b = \bigoplus_{i=1}^{N} \beta_i. \tag{30}$$

Bob does not receive I or the selected point.

The PR relations imply

$$a \oplus b = d \oplus \bigoplus_i u_i v_i = d \oplus e_I. \tag{31}$$

Conditioned on the shared data and the pair x, y, the vector of Alice's PR outputs is uniform on $\{0,1\}^N$; this follows directly from the product of (28). Its parity is fair. Equation (31) then makes both final marginals fair, while their sign-product expectation is

$$\mathbb{E}[\operatorname{sgn}(w_x \cdot \lambda_I) \operatorname{sgn}(y \cdot \lambda_I)] = C_Q(x, y), \tag{32}$$

by the selected-point law. An unbiased binary-output joint distribution has probabilities $(1+ABC)/4$. Hence this branch gives the entire Q, and not merely its correlator. The branch coin and (22) give exactly P_θ for all directions.

All preprocessing and output functions in this protocol are Borel. The shared prior, resource, number of queries and local maps are fixed for θ. Every run queries exactly N_θ independent boxes and returns a pair of output bits whose distribution is P_θ. This proves the theorem.

5 Zeros and denominators

The only denominator in the local response is ρ_x; its zero set was given an explicit fallback in (18). Whenever $\rho_z > 0$, (21) implies $\rho_x \geq c\rho_z > 0$, so the fallback is unused on the positive-density part of the local branch. In the residual branch, $\widetilde{\rho}_x = 0$ has zero probability under its selected-point law for each x. Great circles $y \cdot \lambda = 0$ have zero probability because this law is absolutely continuous with respect to sphere area. The sign convention still defines the wiring on every point.

Every zero of a target joint probability is preserved. For example, conditional on Alice's sign η, the target conditional Bloch vector is r_η/p_η, and its zero outcome occurs at $y = -Br_\eta/p_\eta$. More generally (22) is a convex decomposition with strictly positive weights and nonnegative branches. If a target event has probability zero, both branch probabilities are zero. The finite sampler reproduces Q exactly for that fixed x, so input-dependent kernel null sets do not weaken this conclusion.

6 References

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